Write a Java program to implement the Diffie-Hellman Key Exchange mechanism | Diffie Hellman Key Exchange in Java | Cryptography

AIM: Implement the Diffie-Hellman Key Exchange mechanism using HTML and JavaScript.

HARDWARE AND SOFTWARE REQUIREMENT:

1. Intel based Desktop PC: - RAM of 512 MB

2. Notepad/Notepad ++ editor

3. Net beans / Eclipse

THEORY:

Ø Diffie Hellman (DH) key exchange algorithm is a way for securely exchanging cryptographic keys over internet(Public channel). Keys are not actually exchanged – they are jointly derived.

Ø If 1st Person and 2nd Person want to communicate with each other, they first agree between them a large prime number **P**, and a generator (or base) **G** (where 0 < G < P).

Ø 1st Person select a secret integer "**a**" (his private key) and then calculates $\mathbf{x=G^a \bmod P}$ (which is his public key). 2nd Person chooses his private key **"b'**, and calculates his public key $\mathbf{y=G^b \bmod P}$

Ø 1st Person and 2nd Person exchange values of x and y.

Ø 1st Person and 2nd Person compute symmetric keys by using

$K_a=y^a \bmod P$ and $k_b= x^b \bmod P$

Ø Example:

Step 1 : 1st Person and 2nd Person get public numbers P = 23,G = 9

Step 2: 1st Person selected a private key a = 4 and 2nd Person selected a private key b = 3

Step 3: 1st Person and 2nd Person compute public values

1st Person : $x =(9^4 \bmod 23) = (6561 \bmod 23) = 6$

2nd Person: $y = (9^3 \bmod 23) = (729 \bmod 23) = 16$

Step 4: 1st Person and 2nd Person exchange public numbers

Step 5: 1st Person receives public key y =16 and 2nd Person receives public key x = 6

Step 6: 1st Person and 2nd Person compute symmetric keys

1st Person : $k_a = y^a \bmod p = 65536 \bmod 23 = 9$

2nd Person: $k_b = x^b \bmod p = 216 \bmod 23 = 9$

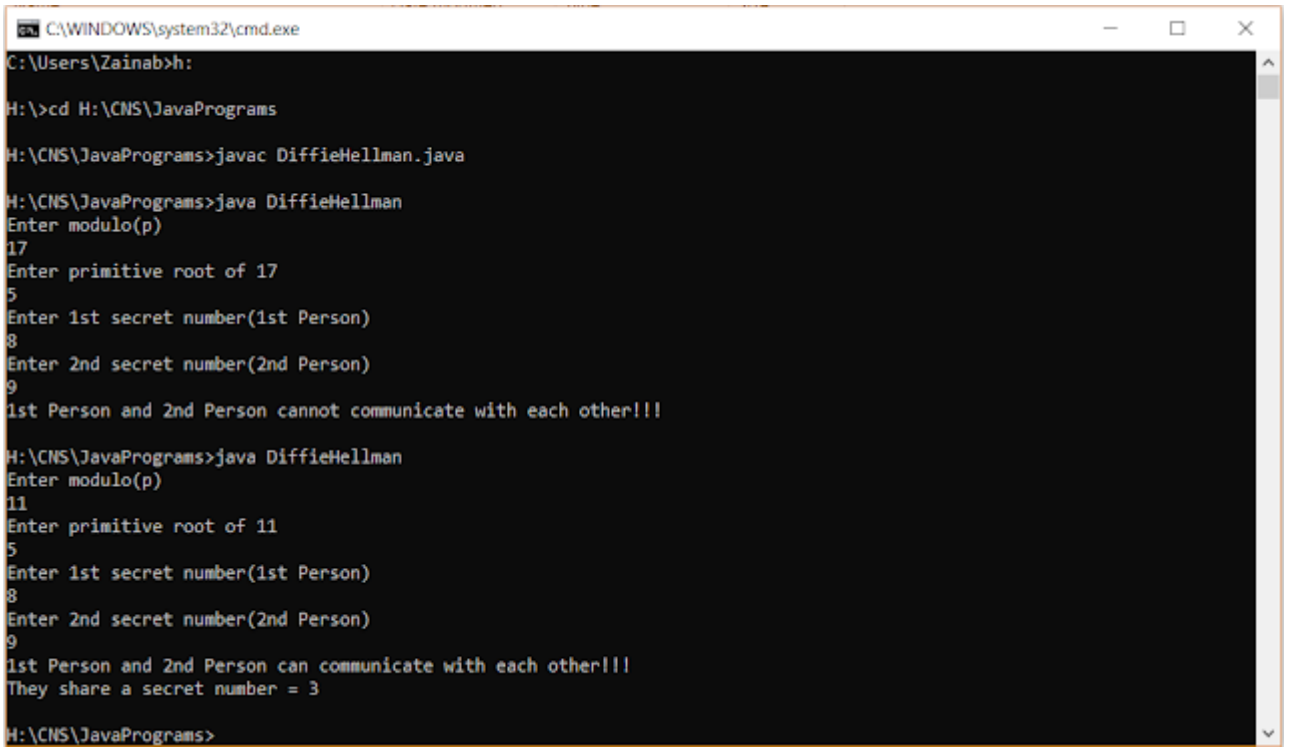
Step 7: 9 is the shared secret

SOURCE CODE:

```
import java.util.*;
class DiffieHellman
{
   public static void main(String[] args)
   {
      Scanner sc=new Scanner(System.in);
      System.out.println("Enter modulo(p)");
      int p=sc.nextInt();
      System.out.println("Enter primitive root of "+p);
      int g=sc.nextInt();
      System.out.println("Enter 1st secret number(1st Person)");
      int a=sc.nextInt();
      System.out.println("Enter 2nd secret number(2nd Person)");
      int b=sc.nextInt();
      int A = (int)Math.pow(g,a)%p;
      int B = (int)Math.pow(g,b)%p;
      int S_A = (int)Math.pow(B,a)%p;
      int S_B =(int)Math.pow(A,b)%p;
      if(S_A==S_B)
      {
         System.out.println("1st Person and 2nd Person can communicate with each other!!!");
         System.out.println("They share a secret number = "+S_A);
```

```
        }
        else
        {
            System.out.println("1st Person and 2nd Person cannot communicate with each other!!!");
        }
    }
}
```

OUTPUT:



VIVA QUESTIONS:

1. What is JAVA?
2. What is Diffie-Hellman Algorithm?
3. What is java script?
4. What is Cryptography?
5. What is Encryption?
6. What is Decryption?